

Lab#02

Department of Computer Science (7th Semester)

Course: DevOps

Name:

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SUBMITTED TO:

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Registration Number

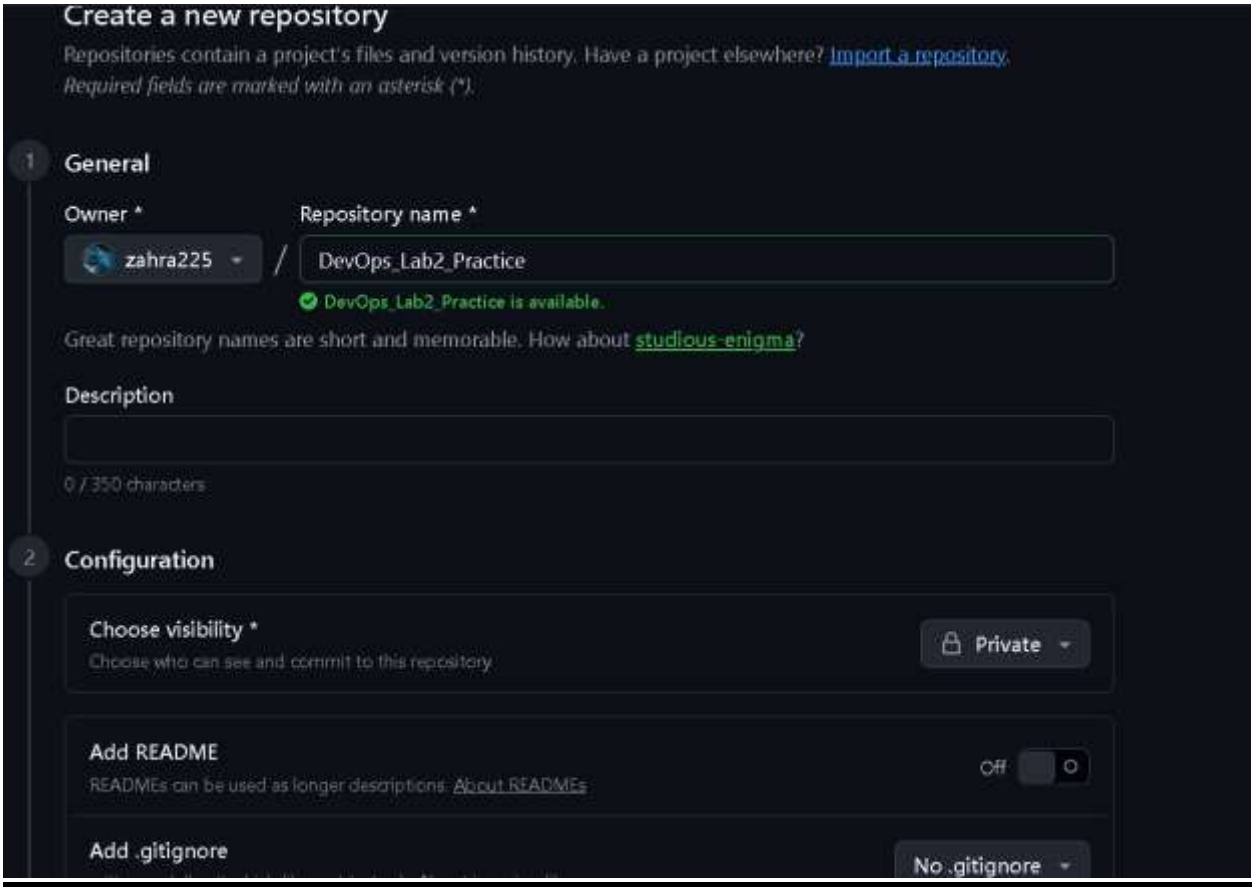
23-BCS-092(Section-A)

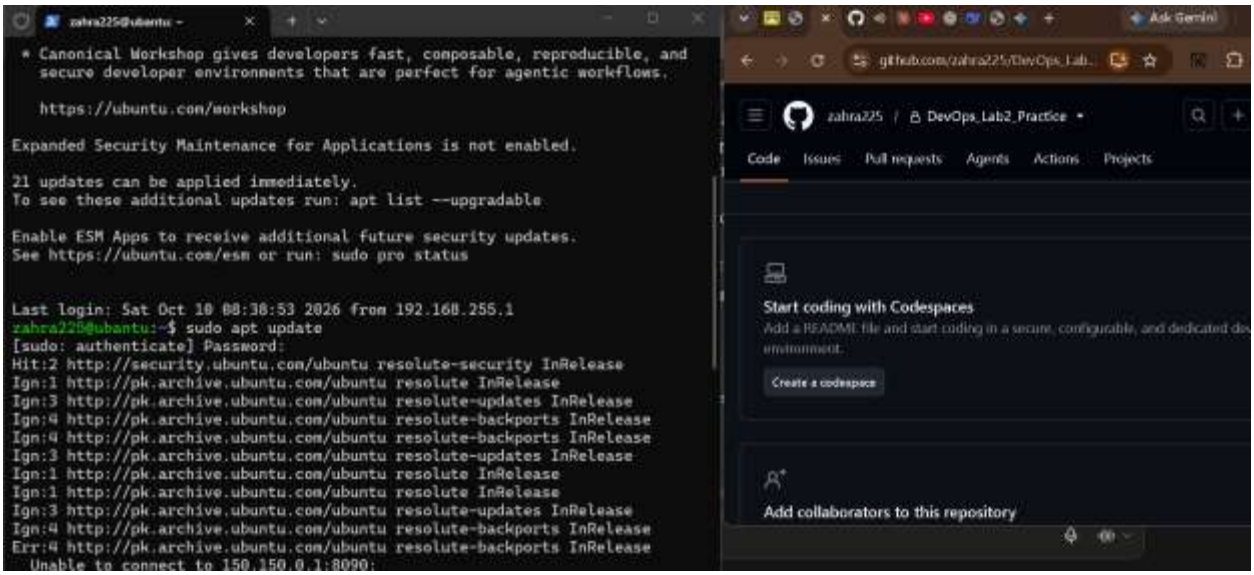
Lab Title:

Hands-On Git & Version Control Lab

Task 1: Create Private GitHub Repository

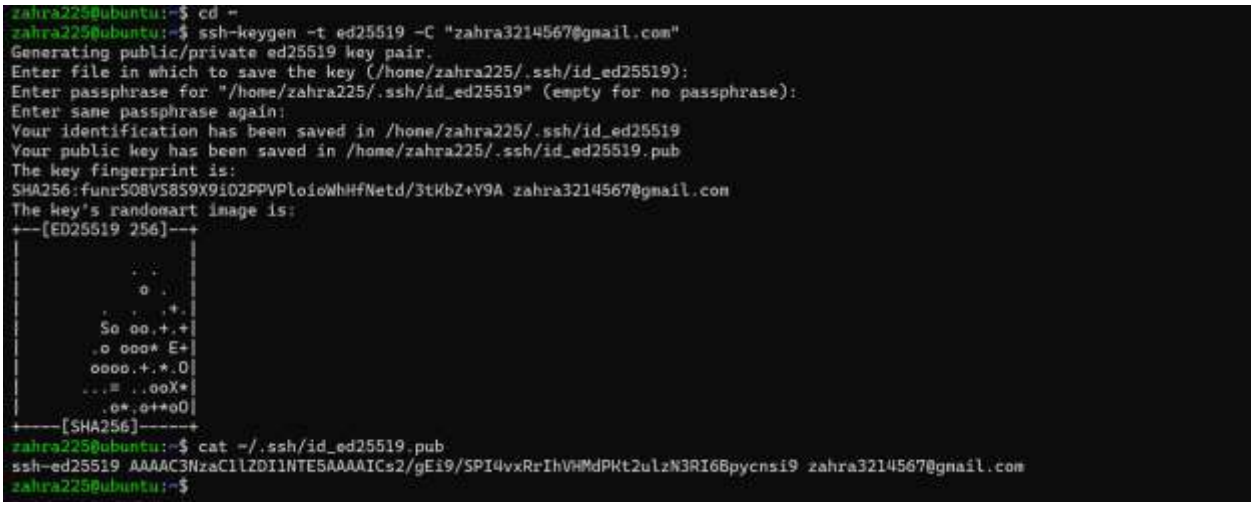
1. Create a new private repository named Lab2 on GitHub.



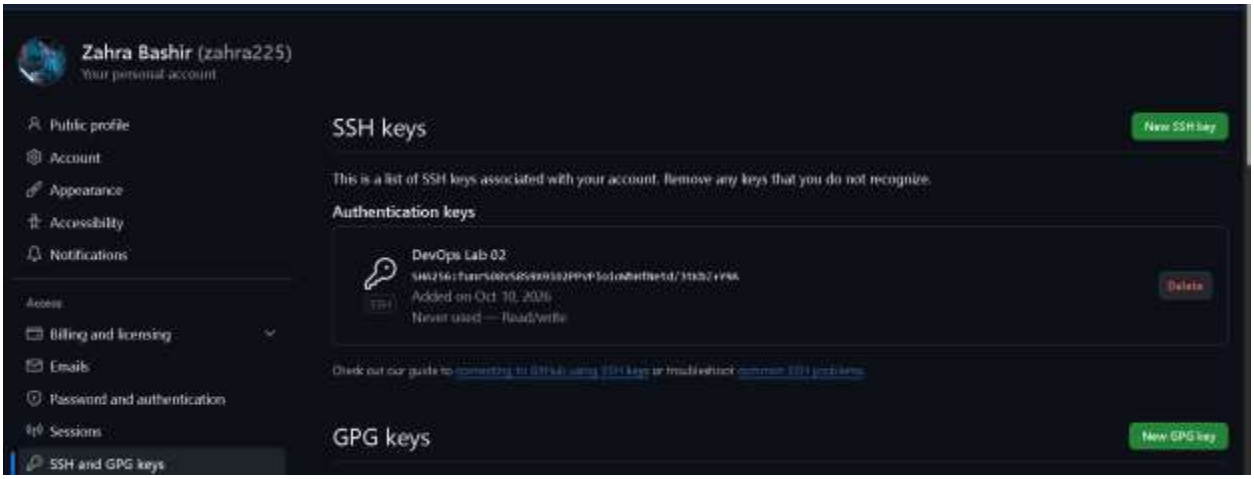


Task 2: Connect Repository via SSH

1. Generate a new SSH key using PowerShell:



2. Add your SSH public key to GitHub (Settings > SSH and GPG keys).



3. Clone your Lab2 repo using SSH.

```
zahra225@ubuntu:~$ git clone git@github.com:zahra225/DevOps_Lab2_Practice.git
Cloning into 'DevOps_Lab2_Practice'...
warning: You appear to have cloned an empty repository.
zahra225@ubuntu:~$ ls
DevOps  DevOps_Lab2_Practice
zahra225@ubuntu:~$ cd DevOps_Lab2_Practice
zahra225@ubuntu:~/DevOps_Lab2_Practice$ pwd
/home/zahra225/DevOps_Lab2_Practice
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

Task 3: Configure Git Username and Email

1. Set up your Git identity (this ensures all commits are linked to you):

```
zahra225@ubantu:~$ git config --global user.name "Zahra Bashir"
zahra225@ubantu:~$ git config --global user.email "zahra3214567@gmail.com"
```

2. Verify your configuration:

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git config --list
user.name=Zahra Bashir
user.email=zahra3214567@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
remote.origin.url=git@github.com:zahra225/DevOps_Lab2_Practice.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

Task 4: Explore the .git Folder

```
GC@Zahra MINGW64 /c/Windows/System32/DevOps_Lab2_Practice (main)
$ ls -a .git
./  ../  HEAD  config  description  hooks/  info/  objects/  refs/

GC@Zahra MINGW64 /c/Windows/System32/DevOps_Lab2_Practice (main)
$
```

Task 5: Local Repository Management

1. Delete the existing .git folder from your cloned repo using Git Bash:

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ ls -a
.  ..  .git
zahra225@ubuntu:~/DevOps_Lab2_Practice$ ls -a .git
.  ..  HEAD  config  description  hooks  info  objects  refs
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

2. Re-initialize the local git repository:

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/zahra225/DevOps_Lab2_Practice/.git/
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

3. Add a file named README.md and commit it:

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ echo "# Lab02 Git Practice" >> README.md
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Initial commit"
[master (root-commit) 605764b] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

4. Connect your local repo to GitHub and push:

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ echo "# Lab02 Git Practice" >> README.md
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Initial commit"
[master (root-commit) 605764b] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git remote add origin git@github.com:zahra225/DevOps_Lab2_Practice.git
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git branch -M main
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 240 bytes | 60.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:zahra225/DevOps_Lab2_Practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

Task 6: File Status & Staging

1. Create a new file notes.txt and write a note. Check status:

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ echo "Adding some test notes" >> notes.txt
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        notes.txt

nothing added to commit but untracked files present (use "git add" to track)
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

2. Stage and commit:

- git add notes.txt
- git commit -m "Add notes.txt"

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ echo "Adding some test notes" >> notes.txt
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        notes.txt

nothing added to commit but untracked files present (use "git add" to track)
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   notes.txt

zahra225@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add notes.txt"
[main d279813] Add notes.txt
 1 file changed, 1 insertion(+)
 create mode 100644 notes.txt
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
d279813 (HEAD -> main) Add notes.txt
605764b (origin/main) Initial commit
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

3. Edit notes.txt and repeat status/commit steps.

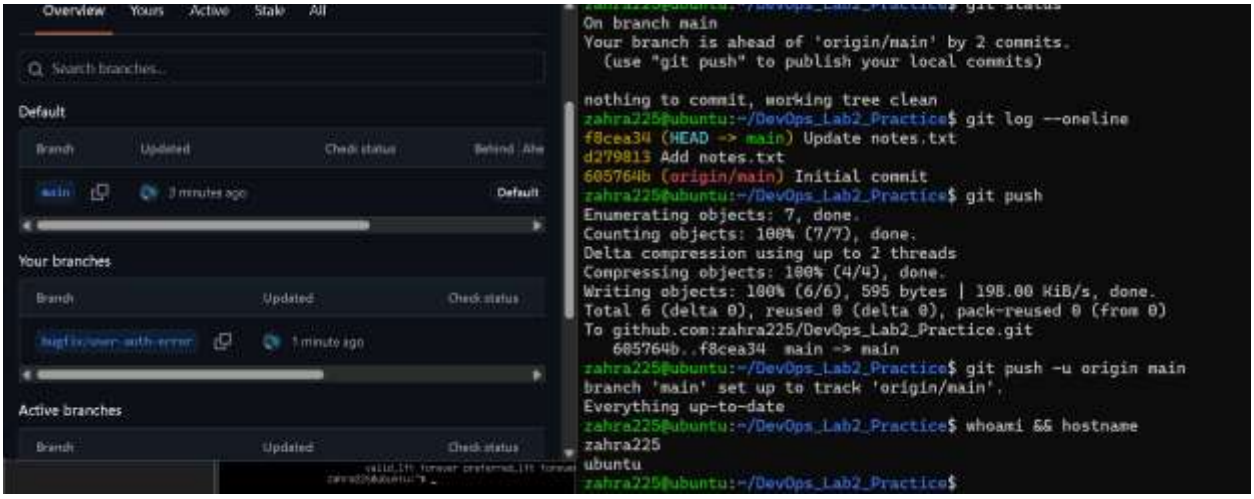
```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ echo "Second line of notes" >> notes.txt
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   notes.txt

no changes added to commit (use "git add" and/or "git commit -a")
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git diff
diff --git a/notes.txt b/notes.txt
index 9f885ab..0807a1e 100644
--- a/notes.txt
+++ b/notes.txt
@@ -1 +1,2 @@
 Adding some test notes
+Second line of notes
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Update notes.txt"
[main f8cea34] Update notes.txt
 1 file changed, 1 insertion(+)
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
f8cea34 (HEAD -> main) Update notes.txt
d279813 Add notes.txt
605764b (origin/main) Initial commit
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

Task 7: Branch Creation Using GitHub GUI

1. On GitHub (web interface), create a branch named bugfix/user-auth-error.



2. Pull the branch to your local repository to sync.
3. git pull origin bugfix/user-auth-error

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
* main
  remotes/origin/bugfix/user-auth-error
  remotes/origin/main
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git checkout bugfix/user-auth-error
branch 'bugfix/user-auth-error' set up to track 'origin/bugfix/user-auth-error'.
Switched to a new branch 'bugfix/user-auth-error'
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git branch
* bugfix/user-auth-error
  main
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

Task 8: Branch Creation and Push Using Git Bash

1. Create a branch named feature/db-connection using Git Bash:

- git checkout -b feature/db-connection
- Push the branch to the remote repository:

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git branch
  bugfix/user-auth-error
* feature/db-connection
  main
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/db-connection
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:      https://github.com/zahra225/DevOps_Lab2_Practice/pull/new/feature/db-connection
remote:
To github.com:zahra225/DevOps_Lab2_Practice.git
 * [new branch]      feature/db-connection -> feature/db-connection
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
  bugfix/user-auth-error
* feature/db-connection
  main
  remotes/origin/bugfix/user-auth-error
  remotes/origin/feature/db-connection
  remotes/origin/main
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

Task 9: Branching & Merging

1. Create and switch to a branch feature-1:

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-1
Switched to a new branch 'feature-1'
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git branch
  bugfix/user-auth-error
* feature-1
  feature/db-connection
  main
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

2. Modify main.py (add a function) and commit.

```
def greet(name):
    return f"Hello, {name}!"

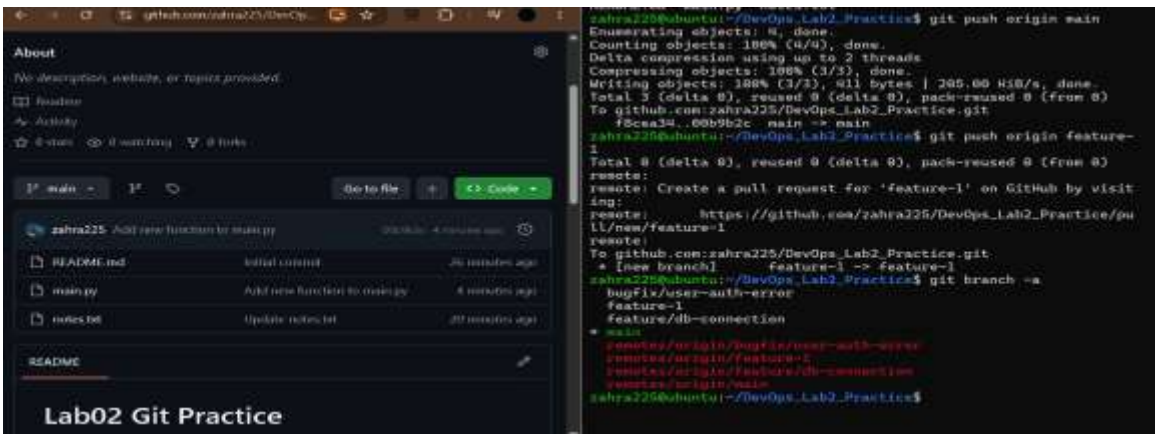

if __name__ == "__main__":
    print(greet("Zahra"))
Hello, Zahra!
On branch feature-1
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        main.py

nothing added to commit but untracked files present (use "git add" to track)
[feature-1 00b9b2c] Add new function to main.py
 1 file changed, 6 insertions(+)
 create mode 100644 main.py
00b9b2c (HEAD -> feature-1) Add new function to main.py
f8cea34 (origin/main, origin/feature/db-connection, origin/bugfix/user-auth-error, main, feature/db-connection, bugfix/user-auth-erro
r) Update notes.txt
d279813 Add notes.txt
605764b Initial commit
```

3. Switch back to main and merge:

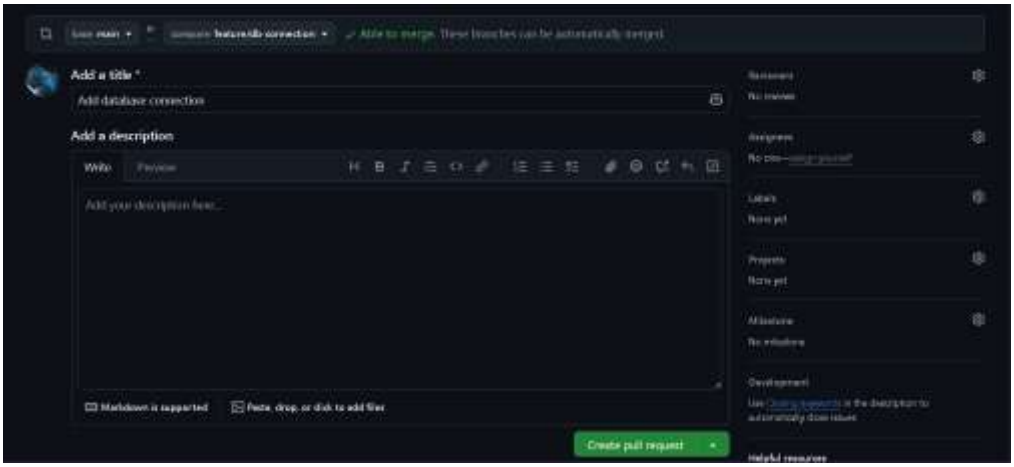
```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git merge feature-1
Updating f8cea34..00b9b2c
Fast-forward
 main.py | 6 ++++++
 1 file changed, 6 insertions(+)
 create mode 100644 main.py
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
00b9b2c (HEAD -> main, feature-1) Add new function to main.py
f8cea34 (origin/main, origin/feature/db-connection, origin/bugfix/user-auth-error, feature/db-connection, bugfix/user-auth-error) Upd
ate notes.txt
d279813 Add notes.txt
605764b Initial commit
zahra225@ubuntu:~/DevOps_Lab2_Practice$ ls
README.md  main.py  notes.txt
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

4. Push all branches:

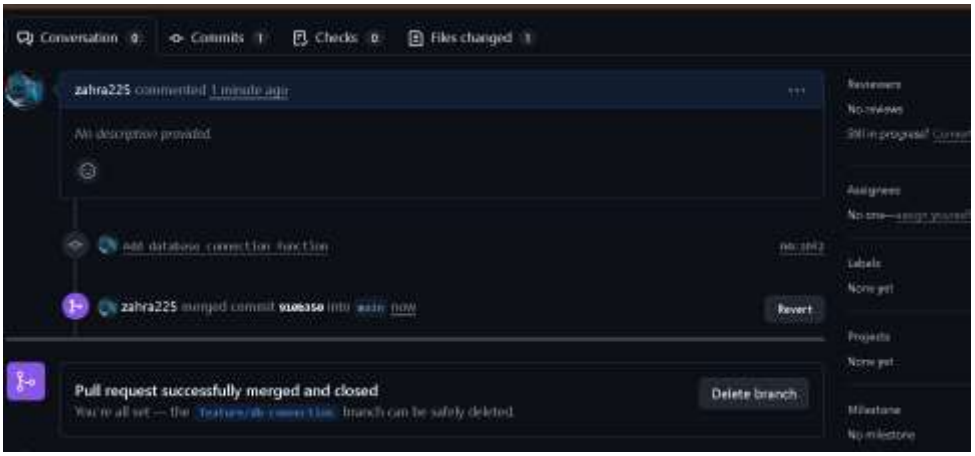


Task 10: Pull Request and Branch Review (GitHub GUI)

1. The Pull Request creation



2. The Pull Request review and merge

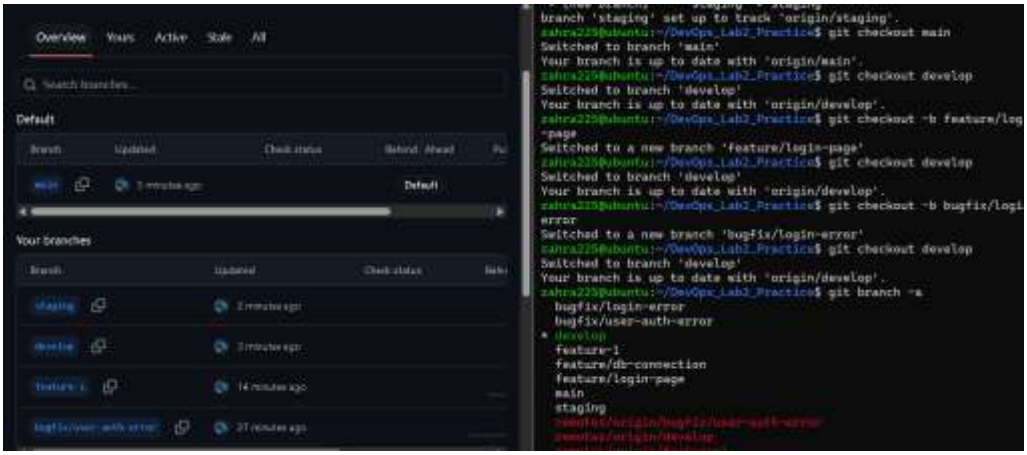


3. The branch deletion



Task 11: Detailed Branch Strategy (Develop/Staging)

1. **Branch list**



2. **Merges into develop and staging**

```
Your branch is up to date with 'origin/develop'.
Already up to date.
Already up to date.
Everything up-to-date
Switched to branch 'staging'
Your branch is up to date with 'origin/staging'.
Already up to date.
Everything up-to-date
*   afdaa7c (HEAD -> staging, origin/staging) Merge develop into staging
|\
| *   77d3199 (origin/develop, develop) Merge bugfix/login-error into develop
| |\
| | * 411e780 (origin/bugfix/login-error, bugfix/login-error) Fix login error handling
| |/
|/|
| * aff09e2 Merge feature/login-page into develop
|/|
| * 0928ed9 (origin/feature/login-page, feature/login-page) Add login function
|/
*   910b350 (origin/main, main) Merge pull request #1 from zahra225/feature/db-connection
|\
| * 04c10f2 (origin/feature/db-connection, feature/db-connection) Add database connection function
* | 00b9b2c (origin/feature-1, feature-1) Add new function to main.py
|/
* f8cea34 (origin/bugfix/user-auth-error, bugfix/user-auth-error) Update notes.txt
* d279813 Add notes.txt
* 605764b Initial commit
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

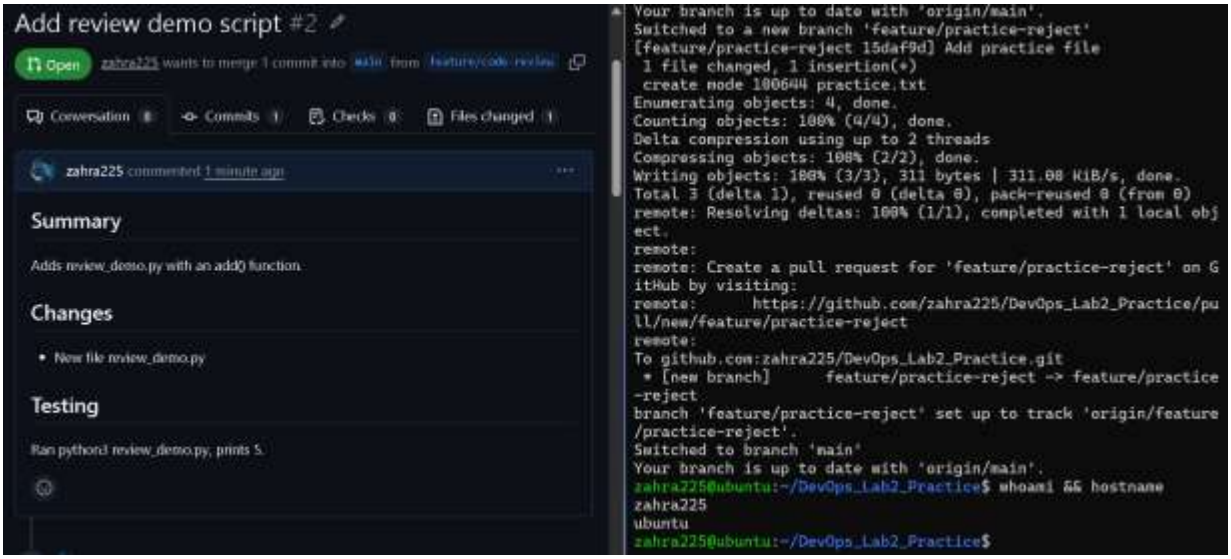
3. **Final merge into main**

```
Your branch is up to date with 'origin/main'.
Already up to date.
Everything up-to-date
*   d19bc70 (HEAD -> main, origin/main) Merge staging into main (release)
|\
| * afdaa7c (origin/staging, staging) Merge develop into staging
|/|
| *   77d3199 (origin/develop, develop) Merge bugfix/login-error into develop
| |\
| | * 411e780 (origin/bugfix/login-error, bugfix/login-error) Fix login error handling
| |/
|/|
| * aff09e2 Merge feature/login-page into develop
|/|
| * 0928ed9 (origin/feature/login-page, feature/login-page) Add login function
|/
*   910b350 Merge pull request #1 from zahra225/feature/db-connection
|\
| * 04c10f2 (origin/feature/db-connection, feature/db-connection) Add database connection function
* | 00b9b2c (origin/feature-1, feature-1) Add new function to main.py
|/
* f8cea34 (origin/bugfix/user-auth-error, bugfix/user-auth-error) Update notes.txt
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```

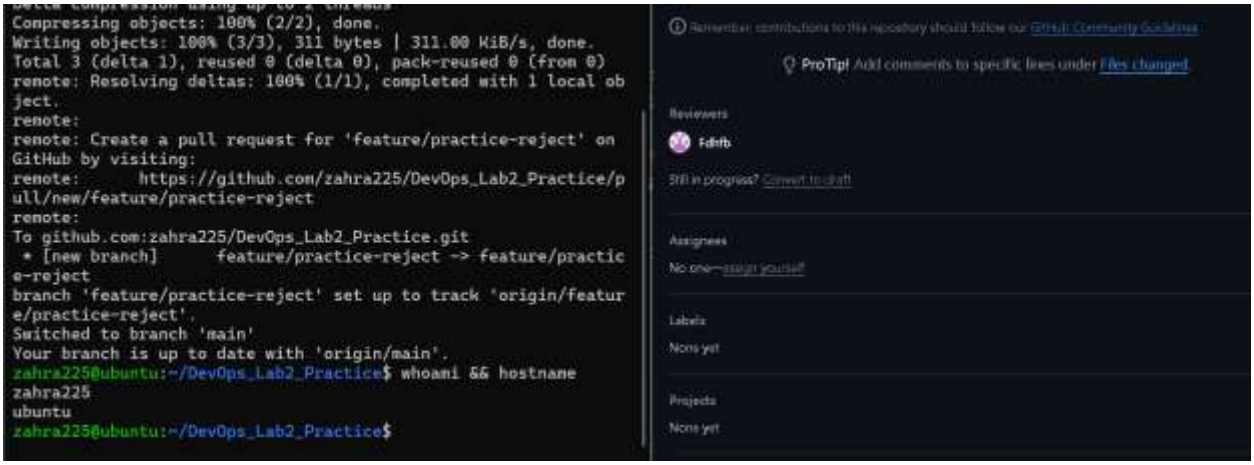
Task 12: Code Review Workflow

1. Create a Pull Request (PR) / Merge Request (MR):

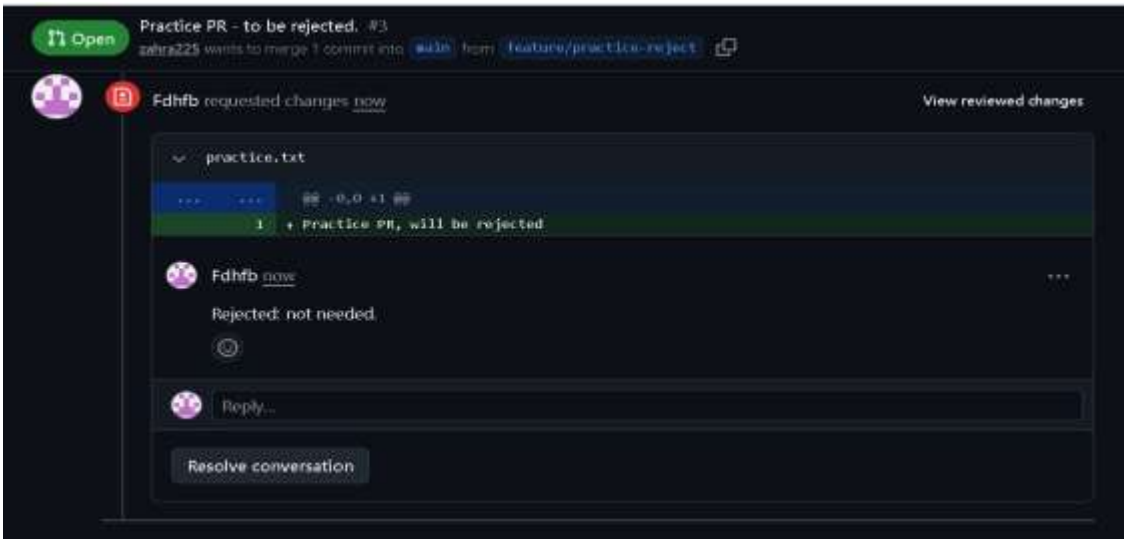
- From a feature branch into main.
- Add a clear title and description summarizing the changes.



2. **Assign a reviewer** (teammate or second account).

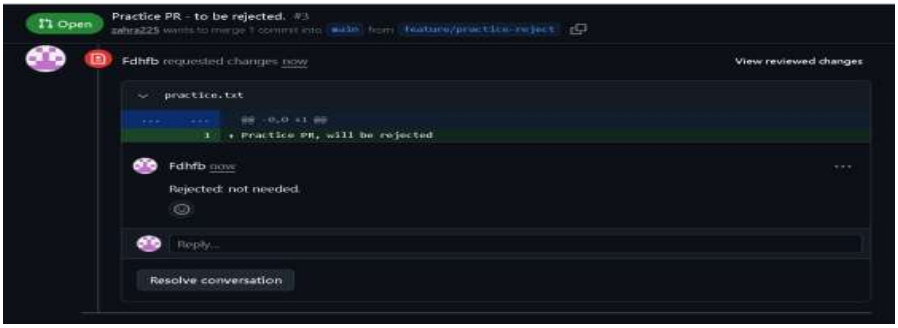


3. **Reviewer Actions (capture screenshots for each):**



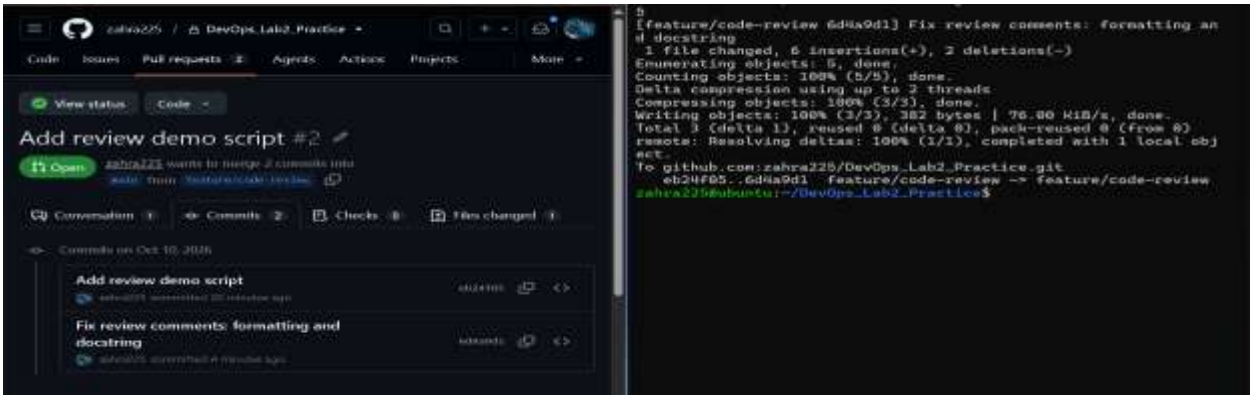
4. **Author Actions:**

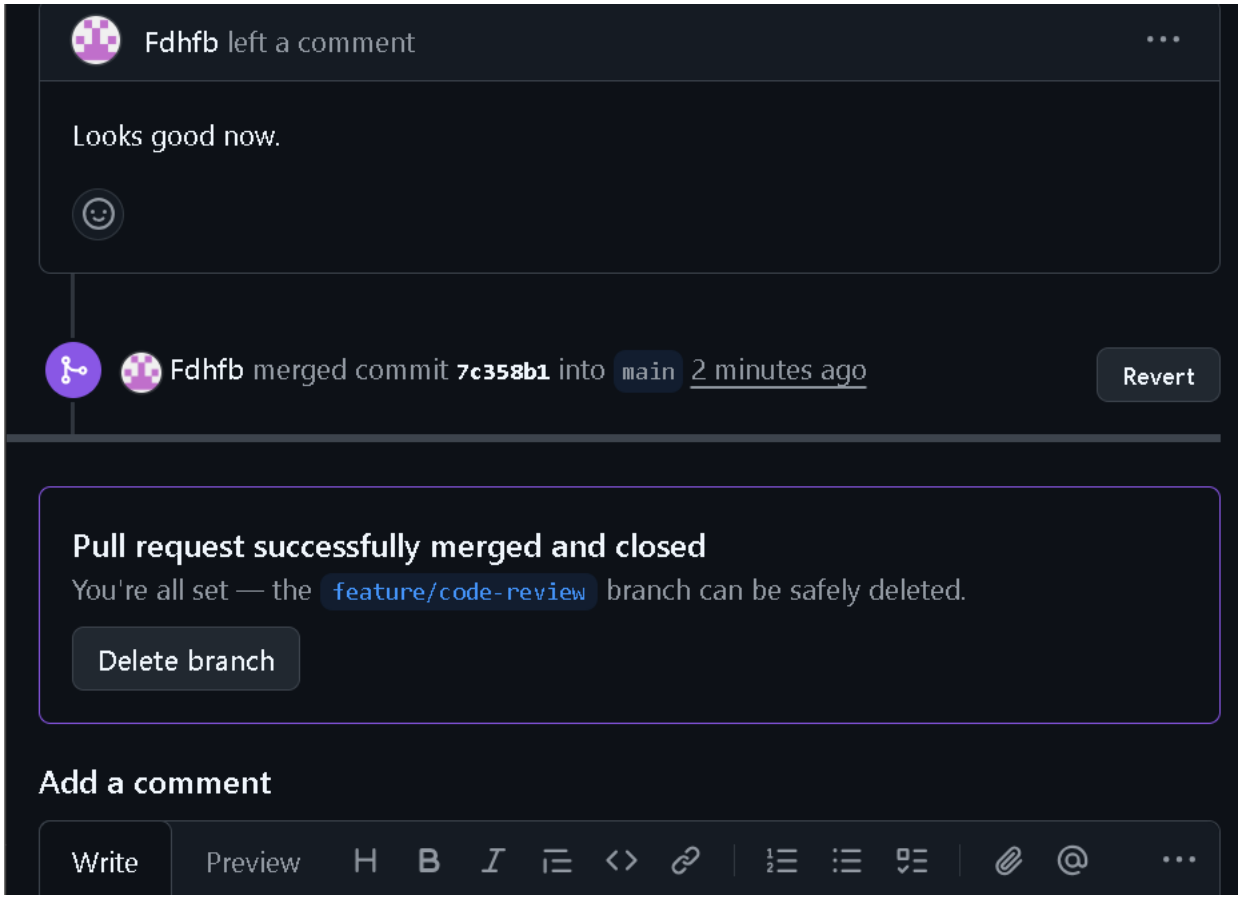
- Fix issues in the same branch and push new commits.
- PR updates automatically.



5. **Merge the PR:**

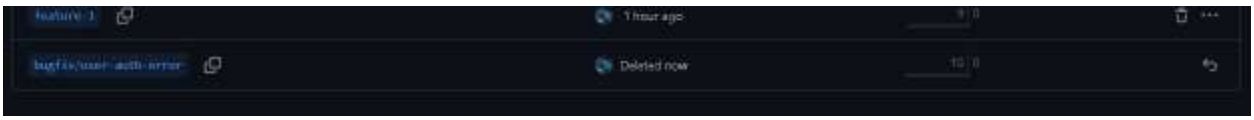
- Reviewer approves → merge using GitHub/GitLab UI.
- Choose merge style (merge commit, squash, or rebase).





6. **Branch Handling After Merge:**

- Delete the merged PR branch

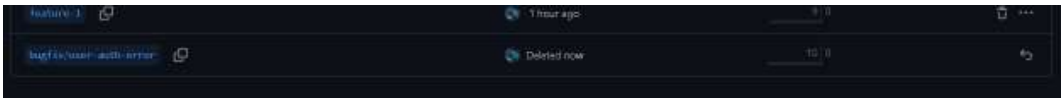


Task 13: Branch Cleanup Best Practices

Delete Remote Branch After Merge:

- Delete a merged branch through the GitHub UI
- Delete another completed remote branch from the terminal:

```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git push origin --delete
 feature-1
To github.com:zahra225/DevOps_Lab2_Practice.git
 - [deleted]         feature-1
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git branch -r
  origin/HEAD -> origin/main
  origin/bugfix/login-error
  origin/bugfix/user-auth-error
  origin/develop
  origin/feature/code-review
  origin/feature/login-page
  origin/feature/practice-reject
  origin/main
  origin/staging
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```



2. **Update Local Repository:**


```
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git push origin --delete
 feature-1
To github.com:zahra225/DevOps_Lab2_Practice.git
 - [deleted]         feature-1
zahra225@ubuntu:~/DevOps_Lab2_Practice$ git branch -r
  origin/HEAD -> origin/main
  origin/bugfix/login-error
  origin/bugfix/user-auth-error
  origin/develop
  origin/feature/code-review
  origin/feature/login-page
  origin/feature/practice-reject
  origin/main
  origin/staging
zahra225@ubuntu:~/DevOps_Lab2_Practice$
```